

Project Status — Behavioral Divergence Between LLM-Driven Multi-Agent Societies and Humans Under Resource Scarcity

Status as of current sprint · Branches: group1-human-study, group2-agents

Group 1 — Sujal & Utkarsh (Human Study)

- Full Streamlit flow: consent (explicit AI co-player disclosure) → instructions → game → debrief
- All six actions implemented, incl. communicate with a structured message slot (kind/value/target/surface)
- session_analysis.py: live debrief analytics — action rates, net resource change, drought-round callout
- logging_utils.py now validates every row against common.schema.validate_row() before writing
- Integrated Group 2's real ScarcityEnv + coplayers: focal-player substitution under a shared seed, decision_latency_ms, pool/co-player-state display for prompt-UI parity
- Full 10-round trial tested end-to-end (incl. a mid-trial co-player death) — passed validate_trial() across all 5 agents
- Co-authored Introduction + Literature Survey, corrected to match the actual built system

Group 2 — Yashash & Saksham (Agents)

- Owns the shared contracts both branches depend on: common/config.py, schema.py, actions.py, ACTIONS.md, LOGGING_SCHEMA.md
- ScarcityEnv: real logistic-regenerating shared pool (not flat per-agent counters) — genuine tragedy-of-the-commons dynamic
- 4 fixed co-player policies (cooperator, free_rider, tit_for_tat, random) behind a frozen act(obs) interface
- RL reflex policy (PPO): 100% vs 10% survival vs. random, 20 held-out seeds
- LLM reasoning layer (Groq + Gemini): live run, 0/50 parse failures; found & fixed 2 real bugs (silent live API calls in tests; Gemini rate-limit misread as model failure)
- Hybrid RL/LLM arbitration: routes social decisions to the LLM, routine actions to RL; 3 arms benchmarked
- Trial campaign infrastructure built (resumable, parallel, manifest-tracked) — real spend not yet authorized
- Ahead of original roadmap: Phases D–F (Weeks 3–5) already complete with real reported numbers

Phase-by-Phase Status

Phase	Status	What was done
A — Contracts	Done	common/config.py, schema.py, actions.py; ACTIONS.md, LOGGING_SCHEMA.md frozen and shared
B — Env + co-players	Done	Logistic-regenerating shared pool; calm/drought/repeated_trust configs; 4 fixed co-player policies; ~20 pilot trials
C — Actions + comms	Done	Six-action space; slotted messages (kind/value/target/surface) make deception arithmetic, not judged
D — RL reflex policy	Done	PPO self-play; 100% vs 10% survival vs. random baseline over 20 seeds
E — LLM reasoning	Done	Groq + Gemini providers; live run 50/50 calls, 0 parse failures; 2 real bugs found & fixed
F — Hybrid arbitration	Done	RL/LLM routing by social trigger; 3 arms tested; caught Gemini rate-limit-vs-quality confound
G — Trial campaign	Infra done, not run	Resumable/parallel runner + manifest built; ~1,860-trial budget planned; spend not yet authorized
H — Validation & freeze	Not started	qa_logs.py, frozen ai_logs_v1, Methodology §4.1–4.2 drafting still ahead
Human app integration	Done	app.py now runs the real ScarcityEnv + co-players under a shared seed (focal-player substitution); logging validated against common.schema; full trial tested end-to-end incl. a mid-trial death
Paper	In progress	Introduction + Literature Survey drafted and corrected to match the actual built architecture; cited against Park et al., Survival Games, DECIDE-SIM

Open Items Needing Team Agreement (not yet resolved)

- Co-player policy ordering (A2–A5) in the human app must be confirmed against run_ai_trials.py's actual seed assignment, or trials aren't truly matched
- No agreed matched-seed list yet between the two branches — flagged as a "never cut" item; blocks real data collection on both sides
- hoard/move remain intentional no-ops on both sides (consistent, not a bug) — open Phase C design call, not urgent